Computational Modeling of Estrogen-Mediated Radioresistance in 3D Breast Cancer Spheroids

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1. BACKGROUND & AIM

Radiotherapy is a key treatment for breast cancer, but tumor-cell response varies. ER-positive breast cancer is particularly relevant because estrogen signaling can promote proliferation. Three-dimensional tumor spheroids also develop spatial oxygen gradients that are not captured by simple 2D models. Mathematical and computational modeling can provide a complementary approach to wet-laboratory experiments by allowing different treatment

AIM • Develop a PhysiCell-based proof-of-concept model to visualize how estrogen, radiation dose, and spatial oxygen distribution influence MCF-7-like breast cancer growth.

2. Materials and Methods

Agent-based simulation using PhysiCell

simulations were developed using **PhysiCell**, an open-source physics-based multicellular simulation framework. Each tumor cell was represented as an individual computational agent with its own position, cell-cycle state, volume, viability, and microenvironmental conditions.
Initial population: 20 cells
MCF-7-like ER-positive phenotype
E2 exposure: 1 nM from Day 7
Radiation: 2 Gy or 4 Gy on Day 7
Simulation endpoint for common comparison: Day

Estrogen-dependent proliferation

$$P_{(E2)} = k_{(E2)} \times P_{(control)}$$

$k_{(E2)}$ = 1.25 (model parameter)
∘P_{E2} represents the proliferation-related transition rate under estrogen exposure,
∘P_{control} represents the baseline proliferation rate,
∘k_{E2} is the estrogen proliferation factor.

Radiation response — Linear-Quadratic model

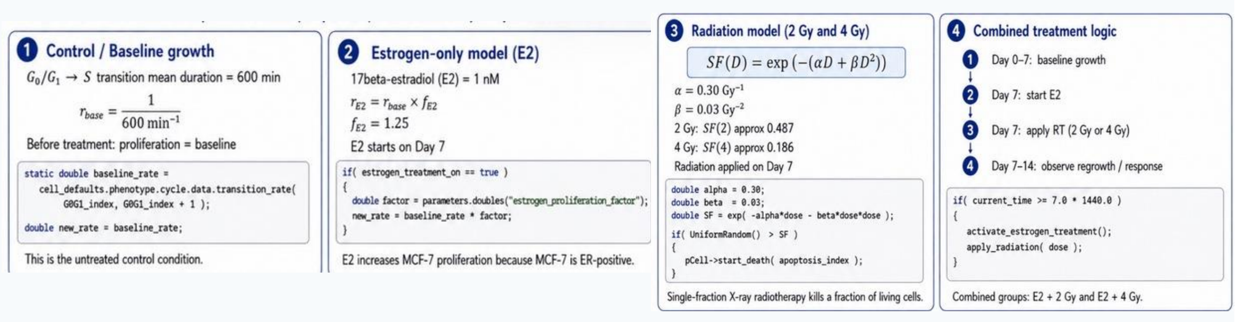
$$SF(D) = \exp[-(\alpha D + \beta D^2)]$$

$\alpha = 0.30\ Gy^{-1}$ $\beta = 0.03\ Gy^{-2}$
∘SF is the surviving fraction,
∘D is radiation dose in Gy,
∘\alpha represents the linear component of radiation injury,
∘\beta represents the quadratic component.

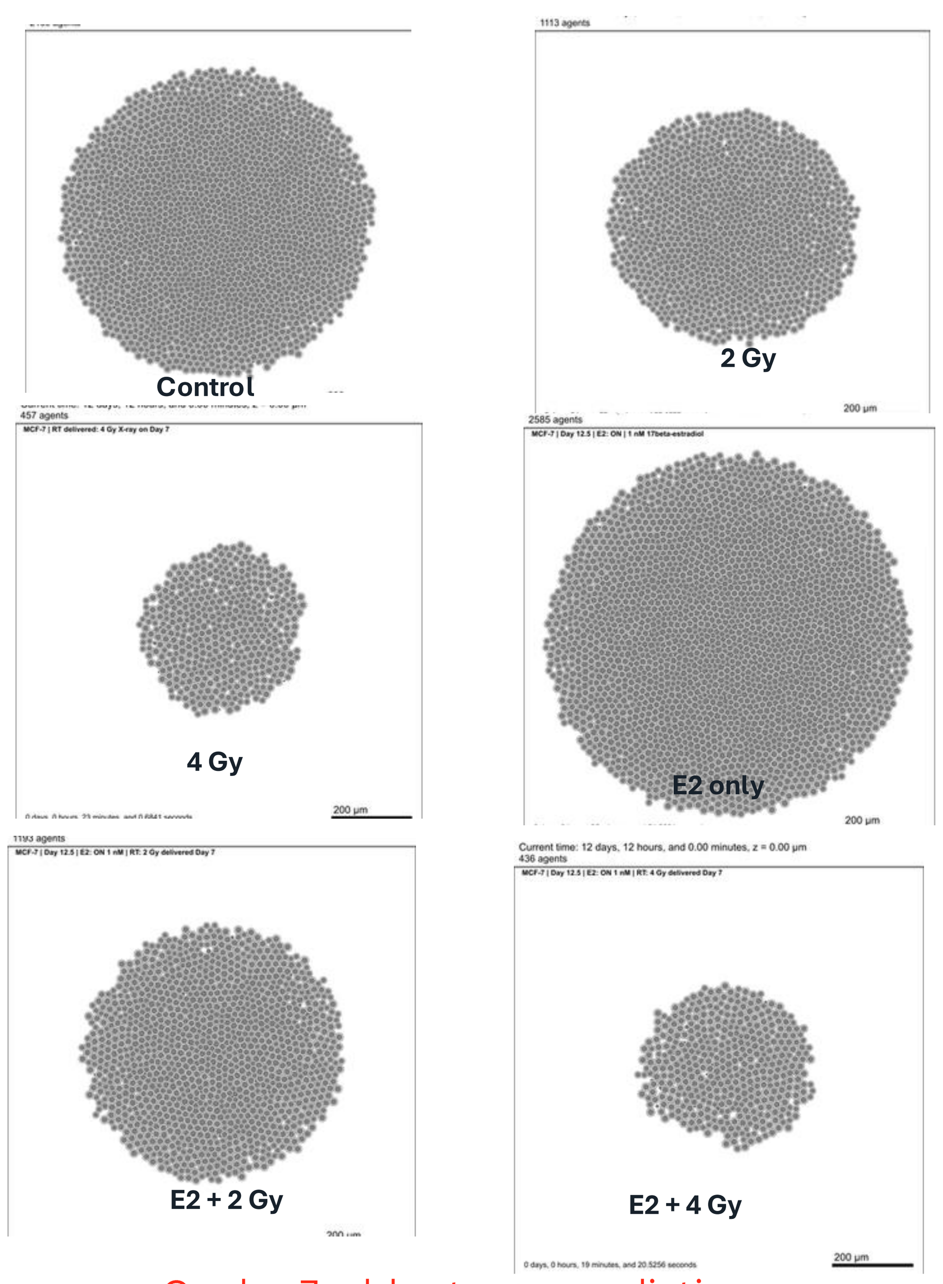
3 SIMULATION DESIGN

Control | E2
2 Gy | 4 Gy
E2 + 2 Gy | E2 + 4 Gy

For combined-treatment simulations, estrogen exposure and irradiation were introduced on day 7.Cell population was recorded throughout the simulation. Day 12.5 was selected as the common quantitative comparison point because values were available for all six treatment groups.All conditions started from the same 20-cell population. E2 and/or radiotherapy were introduced on Day 7.

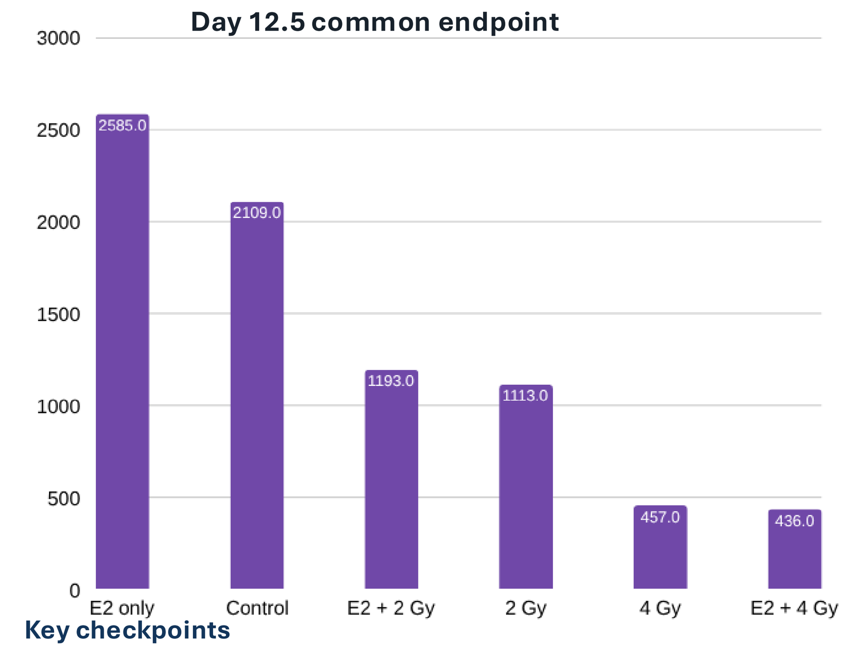


4 2D SIMULATION & SNAPSHOTS



On day 7 add estrogen ,radiation

5 2DSIMULTS —RESULT * CELL POPULATION



Key checkpoints

Group	Day 7	Day 8	Day 12.5
Control	—	—	2109
2 Gy	269	171	1113
4 Gy	244	70	457
E2 only	254	377	2585
E2 + 2 Gy	258	176	1193
E2 + 4 Gy	263	70	436

The untreated control population expanded continuously from the initial 20 cells to 2,109 cells by day 12.5.
Estrogen produced greater simulated growth. The E2-only group contained 254 cells at day 7, increased to 377 cells at day 8 and 573 cells at day 9, and reached 2,585 cells at day 12.5. By day 14, the E2-only simulation contained 4,858 cells.
Thus, among the six conditions investigated, estrogen stimulation produced the highest cell population at the common day-12.5 endpoint.

Radiation suppressed simulated population growth in a dose-dependent manner.
For the **2 Gy group**, the population was 269 cells on day 7 and decreased to **171 cells on day 8**, representing an approximately:
\frac{269-171}{269}\times100
36%reduction.
The surviving population subsequently recovered, reaching **1,113 cells by day 12.5**.
A stronger response was observed with **4 Gy**. Cell number decreased from 244 cells at day 7 to **70 cells at day 8**:
\frac{244-70}{244}\times100
71%.
Despite later regrowth, only **457 cells** were present at day 12.5.
Compared with the untreated control, this corresponded to approximately **47% fewer cells following 2 Gy** and approximately **78% fewer cells following 4 Gy** at day 12.5.

Combined estrogen and radiation treatment
In the **E2 + 2 Gy** simulation, the population decreased from 258 cells at day 7 to **176 cells at day 8**. Subsequent regrowth resulted in **1,193 cells at day 12.5** and **2,233 cells by day 14**.
In the **E2 + 4 Gy** simulation, cell number decreased from 263 cells at day 7 to **70 cells at day 8**, an approximately 73% reduction. Population recovery occurred subsequently, reaching **436 cells at day 12.5** and **831 cells by day 14**.

These simulations therefore demonstrated three principal features: estrogen-associated enhancement of modeled proliferation, radiation dose-dependent population suppression, and subsequent regrowth among surviving irradiated cells.

1.6. OXYGEN / HYPOXIA VISUALIZATION

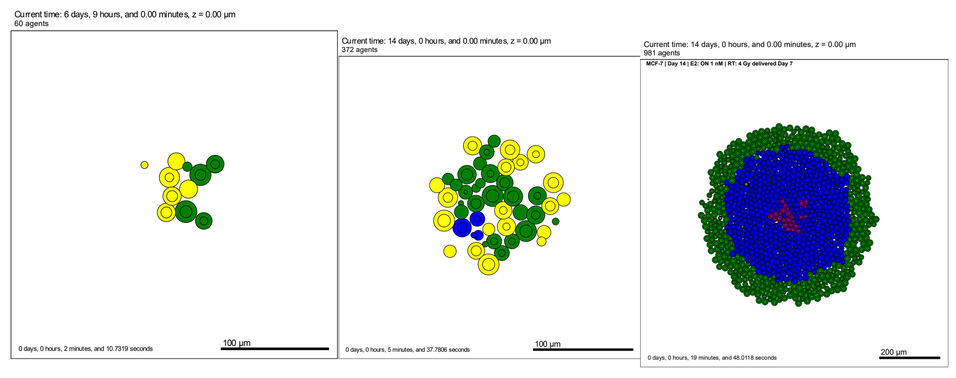
PhysiCell's microenvironment solver was used to simulate oxygen transport in three dimensions.
Oxygen transport can conceptually be described by a diffusion-decay equation:
Oxygen transport was represented with a diffusion–decay framework:

$$\partial C/\partial t = D_{(O_2)} \nabla^2 C - \lambda C - U$$

C = local O_2 concentration • $D_{(O_2)}$ = diffusion • λ = decay • U = cellular consumption

This formulation was included because oxygen within a growing tumor is spatially heterogeneous. Cells located farther from the oxygenated boundary may experience lower local oxygen than peripheral cells.
Cells were color-coded by local simulated oxygen concentration (mmHg) to visualize spatial heterogeneity. The color intervals are visualization bins, not experimentally validated MCF-7 hypoxia thresholds.

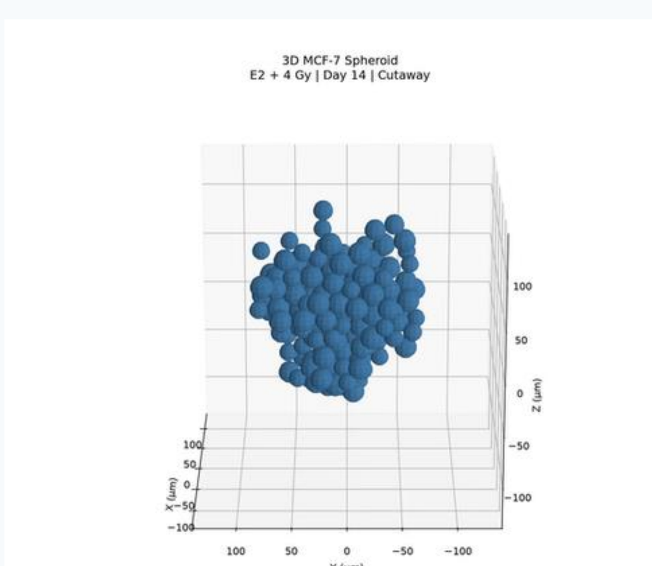
≥35 mmHg | 30–35 mmHg | 20–30 mmHg
10–20 mmHg | 5–10 mmHg | <5 mmHg



1. 2 D to 3D image

Each simulated cell was assigned a true 3D position:

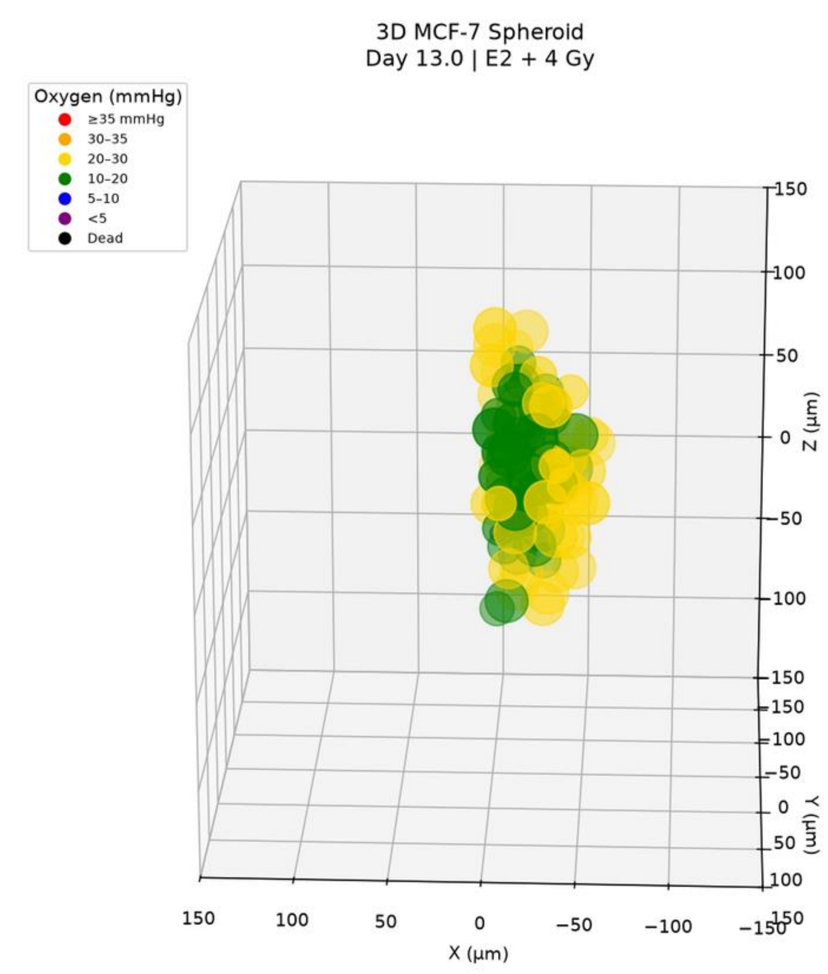
$$x_i = (x_i, y_i, z_i)$$



3D spheroid modeling and cutaway visualization
To extend the model beyond two-dimensional growth, the simulation domain was converted to a true three-dimensional space in which each MCF-7-like cell was assigned independent x, y, and z coordinates. Cell positions were therefore represented as:
\mathbf{x}_i=(x_i,y_i,z_i)
for each individual cell i.

8 OXYGEN3DVISUALIZATION

$$\partial C/\partial t = D_{(O_2)} \nabla^2 C - \lambda C - U$$



The 3D model was used because multicellular tumor spheroids develop spatial organization that cannot be fully represented in two dimensions. In particular, a three-dimensional geometry allows oxygen diffusion and cell crowding to vary throughout the spheroid volume, which is consistent with the project aim of modeling oxygen gradients in a 3D breast cancer microenvironment.

where C is oxygen concentration, D_{O_2} is the oxygen diffusion coefficient, \lambda represents decay, and U represents cellular oxygen consumption.
For visualization, each cell was matched to the oxygen concentration of its nearest microenvironment voxel and colored according to the local oxygen value.

Because the external layers of a full 3D spheroid visually obscure the internal cells, a **half-cut visualization** was introduced. The cutaway was implemented only during rendering by displaying cells on one side of a defined plane, for example:
x_i \leq 0
while cells with
x_i>0
were hidden from the image.

9.KEY FINDINGS and Discussions

The model showed that 4 Gy suppressed cell growth more strongly than 2 Gy, while estrogen increased modeled proliferation. Surviving cells were able to regrow after irradiation, showing that post-treatment growth is also important.
The 3D oxygen visualization also showed spatial differences inside the spheroid, with lower oxygen levels toward internal regions. However, oxygen is currently used mainly for visualization and is not yet directly linked to radiation sensitivity.
Overall, this is a proof-of-concept computational model. Future work should include repeated simulations, parameter testing, and comparison with experimental MCF-7 spheroid data.

1. CONCLUSIONS

Conclusion

Repeated simulations with multiple random seeds
Experimental calibration using 3D MCF-7 spheroids
Sensitivity analysis of α, β, E2 factor and oAn agent-based PhysiCell model was developed to investigate MCF-7-like breast cancer growth under estrogen stimulation and radiotherapy. Estrogen increased modeled proliferation, while 2 Gy and 4 Gy irradiation produced dose-dependent reductions in cell population.
Surviving irradiated cells demonstrated subsequent regrowth, with considerably greater suppression following 4 Gy than 2 Gy.
Extension of the model into three dimensions enabled visualization of spatial oxygen heterogeneity, while a computational cutaway approach allowed internal spheroid structure to be observed.
The current model should therefore be regarded as a proof-of-concept computational framework rather than a validated predictive model. Its principal value is to provide a platform for designing future laboratory experiments and for progressively incorporating experimentally calibrated estrogen signaling, hypoxia, OER-dependent radiosensitivity, and endocrine treatment.

Future work

A further limitation is that the presented values represent computational outputs rather than biological replicates. Statistical significance cannot therefore be inferred from these simulations. Future work should include repeated simulations with different random seeds, calculation of mean values and confidence intervals, sensitivity analysis of parameters such as \alpha, \beta, estrogen proliferation factor, and oxygen uptake, and direct comparison with experimental MCF-7 spheroid data.

11, REFERENCES INFORMATION

AScan the QR code to access **simulation videos, 3D/hypoxia visualizations, project details, results, and references.**

Thanks a lot

SCAN FOR PROJECT DETAILS

Selected references: Hall & Giaccia, Radiobiology for the Radiologist (2018); Friedrich et al., Nat Protoc (2009); Gray et al., Br J Radiol (1953); Ghaffarizadeh et al., PLoS Comput Biol (2018).

TAKE-HOME MESSAGE • Agent-based modeling can generate experimentally testable hypotheses by connecting hormone signaling, radiotherapy, and the 3D tumor microenvironment.